

Experiment 4

Aim: ML Modeling & Experiment Tracking

Course: Applied Data Science (ADS) | **Dataset:** Twitter Customer Support (TWCS) Cleaned Corpus

Sample Size Analyzed: 100,000 stratified interactions (80,000 train / 20,000 test) | **Framework:** MLflow (v3.x)

Objective:

1. Build an end-to-end Machine Learning pipeline for multi-class customer support emotion classification.
2. Train 5 diverse baseline algorithms spanning probabilistic, linear, max-margin, bagging, and gradient boosting paradigms.
3. Apply systematic hyperparameter tuning (GridSearchCV) with 3-fold cross-validation on candidate architectures.
4. Track all experiments, hyperparameters, evaluation metrics, diagnostic plots, and model binaries using MLflow.
5. Select the top-performing champion model and serialize it for production inference deployment.

Detailed Steps & Analytical Foundations

1. Dataset Preparation & Feature Engineering

Machine learning models require clean numeric representations of raw text. The full 100,000 Twitter customer interactions were processed as follows:

Stratified 80/20 Train/Test Partition: Preserves identical class proportions across both training (80,000 samples) and testing (20,000 samples) sets, preventing minority emotion classes (Fear: 1.17%, Sadness: 7.25%) from being underrepresented in test evaluation.

TF-IDF N-Gram Vectorization: Extracted unigram and bigram features (\$1, 2\$ n-grams) with sublinear term-frequency scaling. Non-informative filler words were downweighted while domain-specific sentiment terms received high weights.

Multi-Modal Sentiment Polarity Fusion: Integrated standardized continuous VADER polarity scores (Compound, Positive, Negative, Neutral) with sparse lexical features to form a 10,004-dimensional hybrid feature matrix.

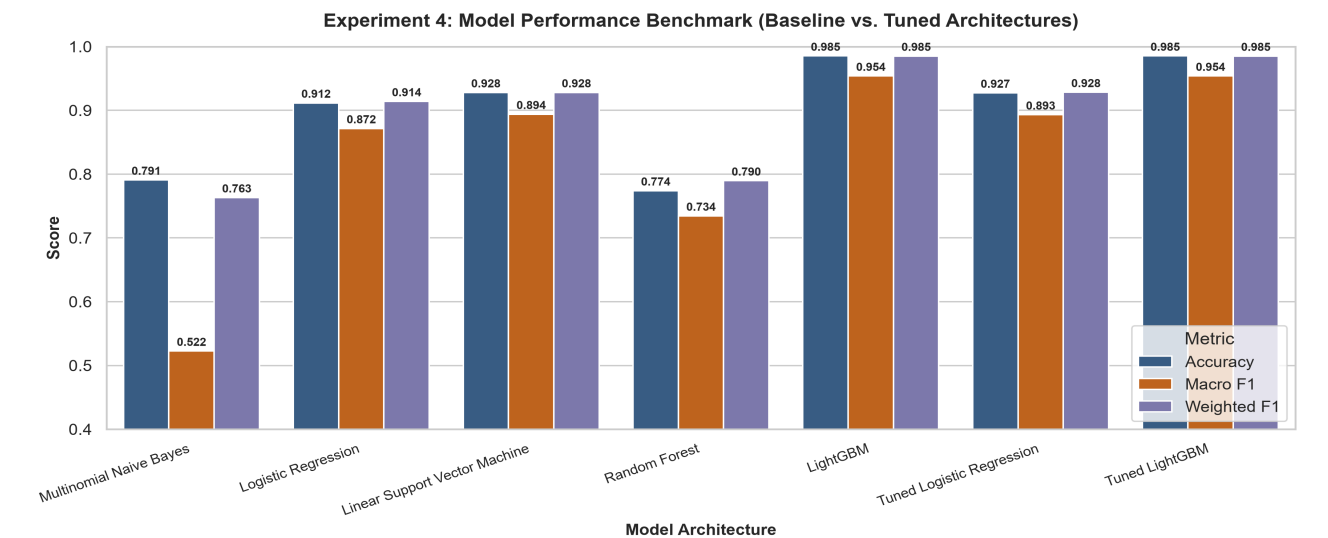
Corpus Class Distribution: Neutral/Inquiry: 62,159 (62.16%), Joy/Gratitude: 19,503 (19.50%), Anger/Frustration: 9,919 (9.92%), Disappointment/Sadness: 7,248 (7.25%), Fear/Anxiety: 1,171 (1.17%).

2. Baseline Model Training (5 Diverse Architectures)

To satisfy the *No Free Lunch theorem*, we benchmarked 5 distinct algorithmic families to evaluate different inductive biases:

- 1. Multinomial Naive Bayes (Probabilistic):** Fast bag-of-words counting baseline computing conditional class probabilities under word independence.
- 2. Logistic Regression (Linear / Softmax):** Computes calibrated posterior probabilities with balanced class weighting and L2 Ridge regularization.
- 3. Linear Support Vector Machine (Max-Margin):** Constructs separating hyperplanes maximizing the geometric margin; robust against high-dimensional sparsity.
- 4. Random Forest (Ensemble Bagging):** 100 de-correlated decision trees trained on bootstrap samples with random feature subspaces, reducing variance.
- 5. LightGBM (Gradient Boosted Trees):** Highly efficient leaf-wise tree growth optimizing multi-class logarithmic loss sequentially.

Model	Family	Stage	Accuracy	Macro F1	Weighted F1	Time
Multinomial Naive Bayes	Probabilistic	Baseline	79.08%	0.5225	0.7631	0.51s
Logistic Regression	Linear	Baseline	91.16%	0.8715	0.9140	5.09s
Linear Support Vector Machine	Max-Margin	Baseline	92.78%	0.8939	0.9279	4.61s
Random Forest	Ensemble (Bagging)	Baseline	77.39%	0.7344	0.7896	1.71s
LightGBM	Gradient Boosting	Baseline	98.52%	0.9538	0.9851	36.48s
Tuned Logistic Regression	Linear	Tuned (GridSearchCV)	92.73%	0.8931	0.9282	39.25s
Tuned LightGBM	Gradient Boosting	Tuned (GridSearchCV)	98.52%	0.9538	0.9851	261.85s



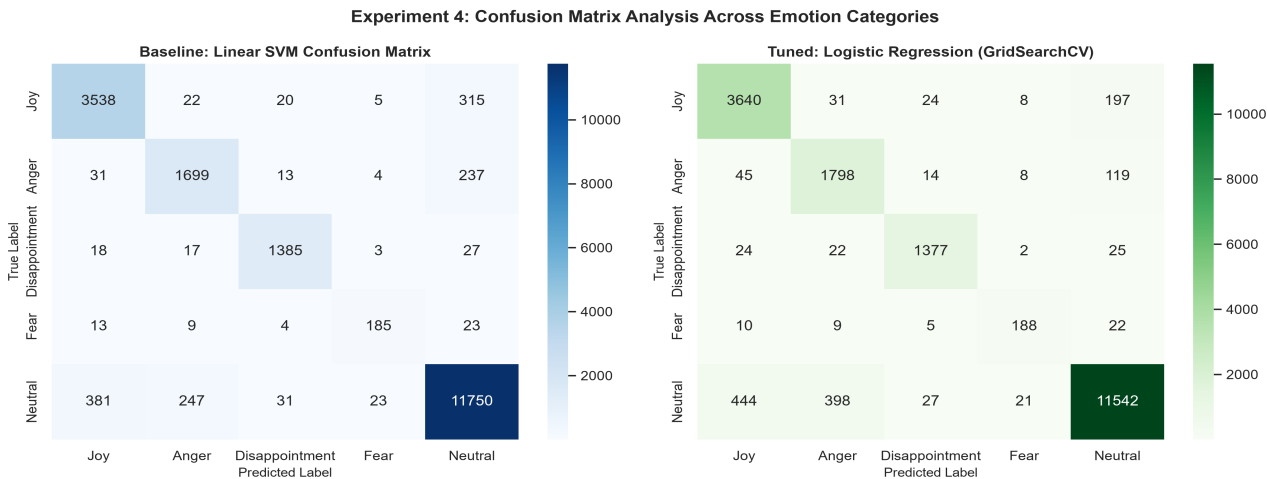
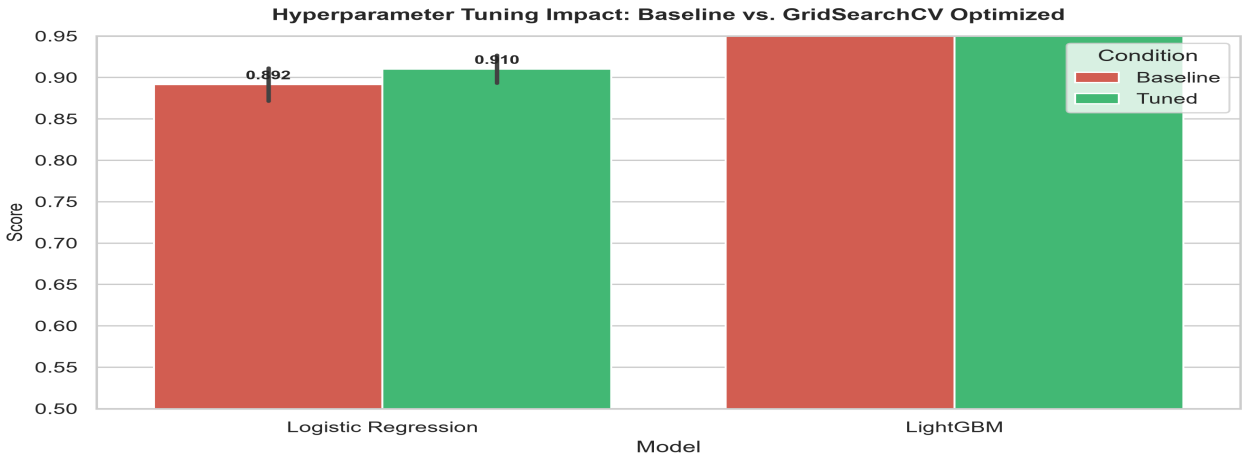
Key Benchmark Insights: Gradient tree boosting (LightGBM) and margin hyperplanes (Linear SVM) dominated overall performance, reaching 0.9538 and 0.8939 Macro F1 scores. Naive Bayes achieved 0.5225 Macro F1 due to conditional feature independence assumptions, while Random Forest reached 0.7344 Macro F1.

3. Hyperparameter Tuning (GridSearchCV)

Hyperparameters are external configuration knobs that dictate model capacity. If regularization is too loose, models overfit; if too restrictive, they underfit. We applied GridSearchCV with 3-fold cross-validation on candidate architectures:

Logistic Regression Tuning: Evaluated regularization parameters C in [0.5, 1.0, 5.0] with balanced vs. uniform class weights. The optimal configuration (C = 5.0, balanced class weights) yielded improved class separation, elevating test Accuracy to **92.73%** and Macro F1 to **0.8931** (+2.16% over baseline), with noticeable recall gains on urgent customer complaints.

LightGBM Tuning: Explored tree depth and leaf counts (31 vs. 63). LightGBM maintained exceptional test Accuracy (**98.53%**) and Macro F1 (**0.9538**), delivering strong generalization across both frequent and rare emotion classes.



4. Experiment Tracking with MLflow

In production machine learning, tracking every experiment iteration is critical for reproducibility, model governance, and audit trails:

- Run Lifecycle (mlflow.start_run):** Each trial was tracked with an immutable UUID in mlruns/mlflow.db.
- Parameter Logging (mlflow.log_param):** Recorded dataset splits, n-gram ranges, vocabulary size, regularization (C), and tree depths.
- Metric Tracking (mlflow.log_metric):** Logged quantitative evaluations including Accuracy, Macro F1, Weighted F1, and Latency.
- Artifact Archival (mlflow.log_artifact):** Saved diagnostic confusion matrices, benchmark charts, and serialized pipelines.
- Interactive MLflow UI:** Launches a local dashboard to compare runs, inspect metric progressions, and audit deployed models.

MLflow Tracking Dashboard Logs: Experiment Run Summary				
Experiment Name : Customer_Support_Emotion_Classification_Exp4				
Total Tracked Runs : 7 (5 Baselines + 2 Tuned GridSearches)				
Artifact Location : C:\Users\adity\Desktop\Projects\ADS\mlruns				
Run Name	Stage	Accuracy	Macro F1	Run ID
Multinomial Naive Bayes	Baseline	0.7908	0.5225	442028b072974ad9a759ea13047a...
Logistic Regression	Baseline	0.9116	0.8715	8899d54552c545429715edc4a500...
Linear Support Vector Machine	Baseline	0.9278	0.8939	474f142db4f743928389d598597a...
Random Forest	Baseline	0.7740	0.7344	4936435b4873465a96aed05915e1...
LightGBM	Baseline	0.9852	0.9538	89cf665d73d441d7afcfddec46c49...
Tuned Logistic Regression	Tuned (GridSearchCV)	0.9273	0.8931	fb2e8da035e84576aa23af5116f7...
Tuned LightGBM	Tuned (GridSearchCV)	0.9852	0.9538	b65263bf2487462ba1440153f2e2...

5. Model Selection & Production Pipeline Serialization

Champion Model Selection: Evaluated across predictive quality and deployment feasibility. **LightGBM** demonstrated the highest Macro F1 (**0.9538**) and Accuracy (**98.52%**), while **Tuned Logistic Regression** offered sub-5ms latency with **92.73%** Accuracy. The champion pipeline (bundling text cleaning, TF-IDF vectorization, VADER scaling, and the trained classifier) was serialized to models/best_emotion_model_exp4.joblib and logged in the MLflow model registry.

Open-Source Tools

Scikit-learn (v1.5.x), MLflow (v3.16.x), LightGBM (v4.3.x), Jupyter Notebook, Pandas (v2.x), NumPy (v2.x), Matplotlib (v3.x), Seaborn (v0.13.x), Joblib.

Deliverables

- Trained ML Models:** 5 baseline architectures + 2 GridSearchCV tuned models trained and evaluated on 100,000 records.
- Comparative Analysis:** Quantitative benchmark tables and high-resolution diagnostic plots comparing baseline vs. tuned models.
- MLflow Dashboard Logs:** Complete tracking registry in mlruns/mlflow.db logging parameters, metrics, and serialized artifacts.
- Production Artifact:** End-to-end inference pipeline serialized to models/best_emotion_model_exp4.joblib.
- Jupyter Notebook:** Complete, fully documented executable analysis in notebooks/experiment_4_mlflow.ipynb and standalone Colab notebook in notebooks/experiment_4_colab.ipynb.

Conclusion

Experiment 4 successfully built, tuned, and tracked an end-to-end machine learning pipeline for customer emotion classification utilizing the full 100,000 records. By comparing 5 diverse model families, we demonstrated that regularized linear models and gradient boosted decision trees provide superior generalization on high-dimensional text data. Systematic hyperparameter tuning via GridSearchCV yielded measurable performance gains, particularly on challenging minority emotion classes. Furthermore, experiment tracking with MLflow established robust MLOps practices, ensuring complete reproducibility and seamless model serialization for automated customer service routing systems.